Introduction to Machine Learning

Learning Paradigms, the Linear Regression Model and the Cost Function

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Week 1 · Part 1

Credits: 3 · —— · Co-req: CSE 816 (Lab)

1. Why machine learning, and where it is used
2. What *is* learning? A formal definition
3. Supervised learning: regression & classification
4. Unsupervised learning
5. The linear regression model
6. The squared-error cost function
7. Cost-function intuition

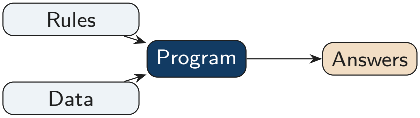
By the end of this session you can

- State Mitchell's definition of learning and instantiate $\langle T, P, E\rangle$ for a given problem.
- Classify a problem as supervised (regression/classification) or unsupervised.
- Write the univariate linear model and its squared-error cost.
- Explain what $J(w, b)$ measures and why we square the errors.

What is Machine Learning?

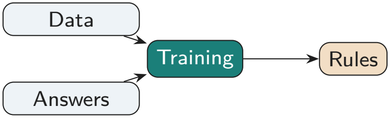
Two ways to make a computer do something

Classical programming



The human supplies the rules. Works when the rules are knowable and writable — sorting, parsing, shortest paths.

Machine learning



The machine *infers* the rules from examples. Works when the rules exist but nobody can state them — recognising a face, pricing a flat, translating a sentence.

Key idea

ML is not magic and not a replacement for algorithms. It is the tool of choice when the specification is available only as *examples*.

Applications you already use today

Consumer scale

- Web search ranking; spam filtering
- Speech recognition, machine translation
- Product / video recommendation
- Photo tagging, OCR of Bangla number plates

Science, industry, public good

- Medical imaging triage; protein structure
- Crop-yield and flood forecasting
- Credit scoring, fraud detection
- Predictive maintenance in manufacturing

Common thread: a large pool of *recorded past decisions* exists, and the decision rule is too intricate or too personal to hand-code.

Defining learning

Arthur Samuel, 1959 — informal

Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.

Tom Mitchell, 1997 — operational

A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

Why the second definition matters. It forces you to name three things before writing a line of code. If you cannot name P, you cannot tell whether your system is working; if you cannot name E, you have no project.

Problem	**Task** T	**Performance** P	**Experience** E
Checkers (Samuel)	Play a game of checkers	Fraction of games won against opponents	Games played against itself
Spam filter	Label an e-mail spam/ham	Fraction correctly labelled	Corpus of user-flagged e-mails
Flat-price estimator	Predict price from size	Mean squared prediction error	Past sale records in Chattogram

Common pitfall

"Accuracy" is not always the right P. For a spam filter that sees 99% ham, a program that always answers "ham" scores 99% and is useless. Choosing P is a design decision — we return to it in Module 2.

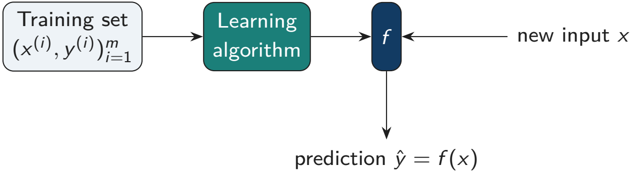
Supervised Learning

Supervised learning: learning from labelled examples

We are given m training examples

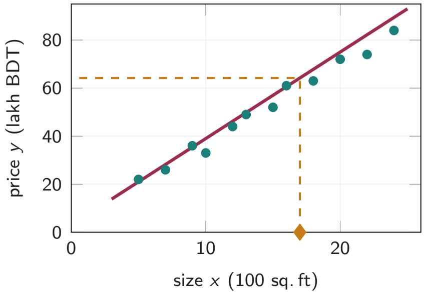
$$\mathcal{D} = \{ (x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)}) \}, \qquad x^{(i)} \in \mathcal{X},\ y^{(i)} \in \mathcal{Y}.$$

The learner must output a function $f : \mathcal{X} \to \mathcal{Y}$ that predicts y well on *new, unseen* x.



The word *supervised* refers to the "right answers" $y^{(i)}$ supplied during training — a supervisor has already solved m instances for us.

Supervised learning, type 1: regression

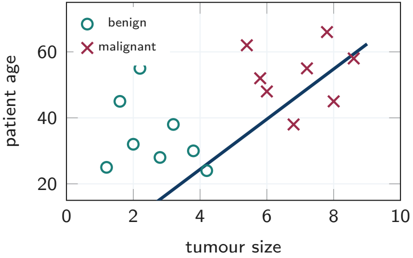


- Infinitely many possible outputs: $\mathcal{Y} = \mathbb{R}$.
- "A 1700 sq. ft flat $\Rightarrow$ about 64 lakh."
- We fit a curve (here, a straight line) and read predictions off it.

Other regression tasks

Forecast tomorrow's rainfall; predict a student's CGPA; estimate remaining battery life; predict delivery time.

Supervised learning, type 2: classification



- $\mathcal{Y} = \{0, 1\}$ (binary) or $\{1, \ldots, K\}$ (multiclass).
- Categories need not be numbers: $\{$benign, malignant$\}$.
- The learner finds a *decision boundary*.

Regression or classification?

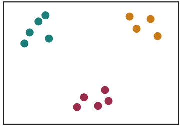
Ask: *is the distance between two outputs meaningful?* Price 52 vs. 53 — yes, nearly the same. Class 1 vs. 2 — no, "cat" is not halfway between "dog" and "bird".

Unsupervised Learning

Unsupervised learning: data with no labels

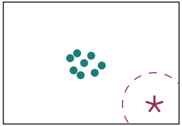
We are given only $\{x^{(1)}, \ldots, x^{(m)}\}$ — no y. The goal is to find *structure* in the data.

Clustering



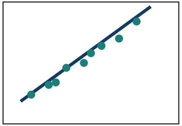
Group similar items: news topics, customer segments.

Anomaly detection



Flag the unusual: fraud, failing engines.

Dim. reduction



Compress many features into a few: PCA.

Key idea

Supervised: "here is the right answer, learn to reproduce it." Unsupervised: "here is the data, tell me something interesting about it."

Quick check: which paradigm?

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   Unsupervised — anomaly detection

The Linear Regression Model

Notation we will use all semester

Symbol	Meaning
x	input variable / *feature*
y	output variable / *target*
m	number of training examples
(x, y)	a single training example
$(x^{(i)}, y^{(i)})$	the i-th training example (i is an *index*, not a power)
$\hat{y}$	the model's prediction, as opposed to the true y
$f_{w,b}(x)$	the model (hypothesis) with parameters w, b

Common pitfall

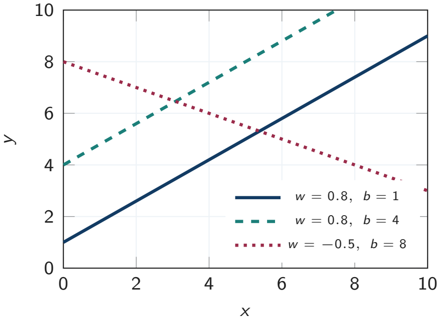
$x^{(2)}$ is the *second training example*; x^2 is x squared. Superscripts in parentheses always index examples.

The univariate linear model

$$\boxed{f_{w,b}(x) \;=\; wx + b}$$

- w — the **weight** or slope: how much $\hat{y}$ changes per unit of x.
- b — the **bias** or intercept: the prediction when $x = 0$.
- w and b are the *parameters*; training means choosing them.
- Called *univariate* linear regression: one input feature.

Shorthand: $\hat{y}^{(i)} = f_{w,b}(x^{(i)}) = wx^{(i)} + b$.



Changing b shifts the line; changing w tilts it.

Why start with a straight line?

Pedagogically

- Only two parameters — we can *draw* the whole optimisation landscape.
- The cost surface is convex: one global minimum, no local traps.
- Every idea here (model, cost, gradient descent) reappears unchanged in logistic regression and neural networks.

Practically

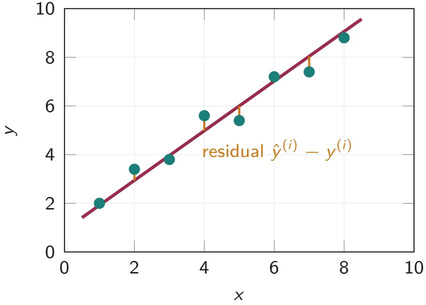
- Still the workhorse of econometrics and epidemiology.
- Interpretable: w has units and meaning.
- A strong baseline — always fit it before a deep network.
- Extends to curves via feature engineering (Week 2).

Key idea

"Linear" means linear *in the parameters* w, b, not necessarily a straight line in x. $f(x) = w_1 x + w_2 x^2 + b$ is still linear regression.

The Cost Function

How do we know one line is better than another?



For each example the model makes an **error** (residual)

$$e^{(i)} = \hat{y}^{(i)} - y^{(i)} = f_{w,b}(x^{(i)}) - y^{(i)}.$$

We want a *single number* summarising all m errors, so that "better" becomes "smaller".

- Sum the raw errors? Positives and negatives cancel — a terrible line could score 0.
- Sum $|e^{(i)}|$? Valid, but not differentiable at 0.
- Sum $(e^{(i)})^2$? Positive, smooth, and penalises large errors more.

The squared-error cost function

Squared-error cost

$$J(w, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{1}{2m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)^2.$$

Reading the formula

- Squaring: either sign costs the same, and a 10-unit miss costs $100\times$ a 1-unit miss.
- Dividing by m: an *average*, so cost does not grow with more data.
- The $\frac{1}{2}$: cosmetic — it cancels the 2 from differentiating.

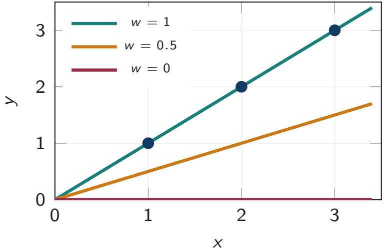
The learning problem

$$\min_{w,\, b} \; J(w, b)$$

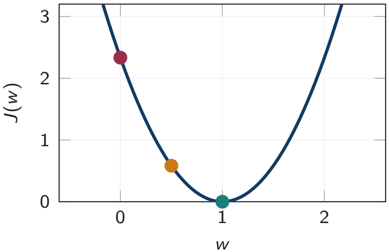
- J is a function of the *parameters*, not of x.
- The data $(x^{(i)}, y^{(i)})$ are fixed constants inside J.
- Training = solving this minimisation numerically — all of Session 2.

Cost-function intuition: a simplified model

Model space: $f_w(x) = wx$ on data $(1,1), (2,2), (3,3)$



Cost space: $J(w)$ as w varies



Each *line* on the left is one *point* on the right. $J(1) = 0$ (perfect fit), $J(0.5) = 0.58$, $J(0) = 2.33$. Minimising J picks $w = 1$.

Worked example — compute J by hand

Data: $(1,1),(2,2),(3,3)$, so $m=3$. Model $f_w(x)=wx$.

$$J(w)=\frac{1}{2\cdot 3}\sum_{i=1}^{3}\left(wx^{(i)}-y^{(i)}\right)^2.$$

At $w=0.5$: predictions 0.5, 1, 1.5; errors -0.5, -1, -1.5.

$$J(0.5)=\frac{1}{6}\left(0.25+1+2.25\right)=\frac{3.5}{6}\approx 0.58.$$

At $w=0$: predictions $0,0,0$; errors $-1,-2,-3$.

$$J(0)=\frac{1}{6}(1+4+9)=\frac{14}{6}\approx 2.33.$$

In general: $J(w)=\frac{1}{6}\left[(w-1)^2+(2w-2)^2+(3w-3)^2\right]=\frac{14}{6}(w-1)^2$ — a parabola with vertex at $w=1$, where $J=0$.

Key idea

J is quadratic in the parameters, so its graph is a parabola (one parameter) or a bowl (two). Convex $\Rightarrow$ a single global minimum.

Wrap-up and Assessment

Concepts

- ML infers rules from data; use it when the rule cannot be written down.
- Mitchell: learning is improvement at T, measured by P, with experience E.
- Supervised = labelled data → regression (continuous y) or classification (categorical y).
- Unsupervised = unlabelled data → clustering, anomaly detection, PCA.

Formulas to memorise

$$f_{w,b}(x) = wx + b$$

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2$$

$$\min_{w,b} J(w, b)$$

Next session

Visualising J as a surface and as contours; the gradient descent algorithm; the learning rate α; gradient descent applied to linear regression.

Descriptive questions

1. State Mitchell's definition of machine learning. For an ML system that learns to detect Bangla handwritten digits, identify T, P and E explicitly.
2. Distinguish between supervised and unsupervised learning. Give one task from agriculture that fits each, and justify your placement.
3. Explain the difference between regression and classification. Why is predicting a house's price a regression problem while predicting whether it will sell within a month is classification?
4. Write the squared-error cost function for univariate linear regression and explain the role of each of the three parts: the square, the factor $1/m$, and the factor $1/2$.
5. For the training set $(1,1),(2,2),(3,3)$ and the model $f_w(x) = wx$, derive a closed-form expression for $J(w)$ and use it to find the minimising w analytically.

6. Argue why the sum of *signed* errors, $\sum_i(\hat{y}^{(i)} - y^{(i)})$, is unusable as a cost function. Support your argument with a concrete two-point data set.
7. "Linear regression can only fit straight lines." Evaluate this statement and justify your verdict.
8. Two students fit the same data. Student A reports $J = 4.2$ using the $\frac{1}{2m}$ convention; student B reports $J = 8.4$ using $\frac{1}{m}$. Have they found different lines? Justify.

Design questions

1. **Framing a problem.** The Chattogram City Corporation wants a system that predicts daily water demand for each ward. Design the formulation: specify $\mathcal{X}$ (what features you would record), $\mathcal{Y}$, the paradigm, and the performance measure P. State two assumptions your design relies on.
2. **Choosing a cost function.** A start-up predicts delivery times. Being 10 minutes *late* is far worse for them than being 10 minutes *early*. Design a modified cost function that encodes this asymmetry, write it mathematically, and state one property (positivity, differentiability, convexity) that your design preserves or breaks.
3. **Auditing a metric.** A hospital's ML screening tool reports 97% accuracy on a data set where 3% of patients have the disease. Explain why the stakeholders should not be satisfied, and design a better evaluation protocol — what would you measure and what data split would you use?
4. **Data design.** You must build the training set for a system that grades short answers in Bangla. Describe how you would collect $(x^{(i)}, y^{(i)})$ pairs, and identify two ways your collection procedure could bias the learned model.

Primary text

G. James, D. Witten, T. Hastie and R. Tibshirani, *An Introduction to Statistical Learning*, 2nd ed., Springer, 2021.

→ **Chapter 2** (§2.1 statistical learning, §2.1.4 supervised vs. unsupervised) and **§3.1** (simple linear regression, least squares).

Supplementary

C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.

→ **§1.1** polynomial curve fitting — the sum-of-squares error function from first principles.

Lab (CSE 816, Week 1): environment setup, model representation and cost-function implementation in NumPy.
Practice quiz: *Supervised vs. unsupervised learning*; *Regression*.